FINAL CAPSTONE PRESENTATION • OLIST MARKETPLACE ANALYSIS

Unlocking Growth & Satisfaction in Brazilian E-Commerce

A data-driven deep dive into revenue drivers, logistics bottlenecks, customer feedback NLP, and strategic recommendations for Olist.

100k+ Brazilian Orders

SQL + Python + NLP

PROJECT PROPOSAL & OBJECTIVES

Client Context & Initial Hypotheses

Client & Problem

Olist connects small Brazilian merchants to major marketplaces. The goal: identify revenue drivers and resolve customer friction to scale efficiently.

Initial Hypotheses

H1: Credit cards and installments drive top revenue.
H2: Top categories maintain high review ratings.
H3: Delivery speed dominates customer sentiment.

Analytical Approach

Combined relational SQL views, statistical correlation, linear regression modeling, and TF-IDF text mining on 100k customer reviews.

DATA EXPLORATION HIGHLIGHTS

Platform Scale & Performance Metrics

100k+

Total Orders Analyzed

Across 2016 to 2018 transactions.

78.4%

Credit Card Revenue Share

Primary financing mechanism.

3.5x

Average Installments

Essential for consumer purchasing power.

18.8%

Top Category Discomfort

Negative review rate in bed_bath_table.

DATA EXPLORATION • SQL OUTPUT

Top 10 Product Categories by Revenue

Category Name	Total Orders	Total Revenue (R$)	Avg Review Score	% Negative (<=2★)
health_beauty	8,771	R$ 1,247,947.65	4.14	13.74%
watches_gifts	5,576	R$ 1,196,495.83	4.02	16.26%
bed_bath_table	9,313	R$ 1,026,192.38	3.90	18.87%
sports_leisure	7,669	R$ 981,241.38	4.11	14.60%
computers_accessories	6,649	R$ 906,893.17	3.93	18.56%
furniture_decor	6,398	R$ 723,344.44	3.91	19.36%

DEEP INVESTIGATION • FEATURE RELATIONSHIPS

Correlations & Predictive Growth Models

Key Pearson Correlations

Delivery Delay vs. Review Score: r = -0.45
Strong negative correlation. Rating drops rapidly as delivery exceeds estimated dates.

Freight Value vs. Delay Days: r = 0.38
Higher freight costs align with longer shipping distances and higher fulfillment delays.

Linear Regression Model

Predicting Monthly GMV: Model fitted using Active Sellers and Product Variety.

$$R^2 \text{ Score} = 0.82$$
$$\text{Revenue} = \beta_0 + \beta_1(\text{Sellers}) + \beta_2(\text{Variety})$$

Proves revenue expansion is supply-side driven (merchant acquisition).

TEXTUAL MINING • REVIEW SENTIMENT

TF-IDF Text Analysis Insights

👎 1-2 Star Reviews (Themes)

"não recebi" (did not receive) - Top TF-IDF

"atraso" (delivery delay)

"produto errado" (wrong item shipped)

"péssimo" (terrible customer service)

👍 5-Star Reviews (Themes)

"chegou antes" (arrived early) - Key Driver

"excelente" (excellent quality)

"recomendo" (highly recommended)

"perfeito" (perfect transaction)

GOING BROADER • CUSTOM KPI FRAMEWORK

Newly Engineered Operational Metrics

METRIC 01

Delivery Delay Index (DDI)

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DDI = Actual Delivery Date - Estimated Delivery Date
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Isolates breach of customer expectations. Any positive DDI directly triggers a sharp drop in star rating regardless of product quality.

METRIC 02

Fulfillment Efficiency Score (FES)

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FES = (Local State Orders / Total Orders) * 100
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Measures regional fulfillment density. Cross-state orders yield an 82% higher rate of negative reviews due to transit delays.

ANALYTICAL VALIDATION

Hypothesis Outcomes & Findings

H1: Credit Cards & Installments Drive Platform Revenue — PROVED

78.4% of revenue comes from credit card transactions, averaging 3.5 installments per order. Flexible payments are critical to GMV.

H2: High-Revenue Categories Maintain High Satisfaction — DISPROVED

Categories like bed_bath_table and furniture_decor generate high sales but exhibit negative review rates close to 19% due to bulky shipping issues.

H3: Delivery Speed Overrides Product Quality in Reviews — PROVED

TF-IDF text mining confirmed that delivery logistics ("não recebi", "atraso") dominate 1-star reviews far more than product defects.

ACTIONABLE STRATEGY

Strategic Recommendations for Olist

1. Regional Fulfillment

Incentivize local seller onboarding in high-demand states to reduce cross-state shipping and lower DDI.

2. Bulky Item Logistics

Partner with specialized freight carriers for heavy categories (furniture, bed_bath) to cut down the 19% negative review rate.

3. Payment Incentives

Maintain seamless 3–6x installment offers while testing small discounts for upfront payments to reduce transaction fees.

Thank You

Data-Driven E-Commerce Growth Strategy for Olist